Yiling Xu

annaxu671@gmail.com 206-861-6637 LinkedIn GitHub Portfolio

Education

New York University Aug 2025 – May 2027
Master of Science in Data Science *New York, New York*
- GPA: 4.0/4.0

University of Washington Oct 2020 – Aug 2024
Bachelor of Science in Informatics, Data Science Track *Seattle, Washington*

Industrial Experience

ByteDance | *Machine Learning Engineer Intern* Jul 2026 – Sep 2026
- Developed an end-to-end multimodal LLM agent integrating video frames, ASR transcripts, titles, and metadata to classify livestream content into 20+ categories and 100+ fine-grained labels.
- Engineered modular prompts incorporating hierarchical classification rules, decision criteria, and counterexamples; improved offline classification accuracy from 71% to 94.6% through iterative prompt optimization.
- Built a scalable batch inference and offline evaluation pipeline with parallel processing, majority voting, and confidence-based review; analyzed 2,600+ human-labeled samples using precision, recall, and class-level error analysis.

Jutze Intelligence Technology | *Machine Learning Engineer Intern* May 2026 – Jul 2026
- Fine-tuned a BiSeNet-based semantic segmentation model for automotive surface defect detection, reducing missed defects by 10% compared with the baseline checkpoint.
- Developed an image tiling and preprocessing pipeline that expanded 37 high-resolution annotated images into 6,924 training samples while preserving small defect boundaries. Reduce missed defects by 10% versus the original checkpoint.
- Evaluated segmentation performance using mIoU, class-wise IoU, precision, recall, and visual diagnostics to identify false positives and missed defects.

Tineco | *Machine Learning Engineer Intern* May 2025 – Jul 2025
- Fine-tuned a BERT-based text classification model to distinguish device faults, user intents, and false triggers from noisy, real-world voice commands.
- Processed and standardized 8,000+ user voice-command records to support model training and improve classification robustness to ambiguous inputs.
- Developed a reproducible NLP pipeline for text preprocessing, model training, evaluation, and batch inference to support fault analysis and experimentation.

Jiangsu EKX Intelligent Technology | *Machine Learning Engineer Intern* Jun 2023 – Sep 2023
- Built reusable AWS Redshift SQL pipelines to clean, join, and validate 150K+ operational records with reconciliation and anomaly checks.
- Centralized KPI definitions in modular SQL transformations and migrated fragmented Tableau dashboards to Amazon QuickSight, eliminating duplicated business logic and improving metric consistency across teams.
- Published curated datasets and self-service dashboard views for recurring operational analysis, reducing repeated manual preparation and accelerating stakeholder reporting workflows.

Research Experience

NYU Immersive Computing Lab | *Research Assistant* Apr 2026 – Present
- Architected and prototyped a selective video-correction system that detects perceptually salient artifacts and inpaints only affected regions, avoiding costly full-video regeneration.
- Created a benchmark spanning AI-generated and real-world distortions to evaluate artifact localization, correction quality, and robustness across diverse failure modes.
- Prototyped gaze-aware salience scoring with efficient masked video inpainting, concentrating computation on perceptually important regions for low-latency enhancement.

Carnegie Mellon University | *Research Assistant* Feb 2023 – Dec 2023
- Designed and automated a benchmarking framework for Fully Homomorphic Encryption schemes, comparing security, runtime, ciphertext expansion, memory, and accuracy across parameter configurations.
- Led the comparative analysis and authored the resulting first-author paper, published at ICDSE 2024.

Projects

NYU VIP: Machine Learning for Energy Commodity Markets | *Python, PyTorch, XGBoost* Jan 2026 – Present
- Implemented XGBoost, LSTM, and Transformer models for forecasting energy commodity prices from historical time-series data.
- Built a unified experimentation framework with Pandas, scikit-learn, and PyTorch for feature generation, temporal splits, training, and model comparison; ran weekly ablations on features, sequence lengths, and error cases.

Skills

Languages: Python, SQL, Java, C++, R, MATLAB, JavaScript/TypeScript, HTML, LaTeX
ML & Data: PyTorch, Pandas, NumPy, Multimodal LLMs, LLM Agents, Fine-tuning, Transformers, BERT, NLP
Tools & Analytics: Git, Linux, Jupyter, AWS Redshift, Weights & Biases, MySQL, MongoDB, Tableau, Amazon QuickSight